

JAEHYEOK BAE

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EDUCATION

Stanford University

Ph.D. in Electrical Engineering

Sep 2024 - Present

Stanford, CA

Seoul National University

B.S. in Electrical and Computer Engineering

Mar 2016 - Aug 2023

Seoul, South Korea

· Graduate *Summa cum laude*, Total GPA : 4.23 / 4.30 (Major GPA : 4.27 / 4.30)

SELECTED PUBLICATIONS

Conference Abstracts and Presentations

- [1] **J. Bae**, Z. Shah, C. Alkan, S. Vasanaawala, J. M. Pauly, and K. Setsompop, “AdaSamp: Towards simple, subject-specific Adaptive Sampling for 3D Accelerated MRI,” in *Proceedings of the ISMRM Annual Meeting*, 2026. **Oral; Summa Cum Laude Merit Award.**
- [2] **J. Bae**, C. Alkan, S. Vasanaawala, J. M. Pauly, and K. Setsompop, “QuickSamp: Towards simple, real-time-optimized Sampling Patterns for 3D Accelerated MRI,” in *Proceedings of the ISMRM Annual Meeting*, 2025. **Oral; Magna Cum Laude Merit Award.**

Conference and Workshop Papers

- [3] K. Kim and **J. Bae** et al., “3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation,” in *IEEE International Symposium on Biomedical Imaging*, 2026
- [4] S. Hong, **J. Bae**, J. Lee, and S. Y. Chun, “Adaptive Selection of Sampling-Reconstruction in Fourier Compressed Sensing,” in *Proceedings of The European Conference on Computer Vision (ECCV)*, 2024
- [5] **J. Bae**, J. Lee, and S. Kim, “PNI : Industrial Anomaly Detection using Position and Neighborhood Information,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023
- [6] J. Kim, **J. Bae**, G. Park, D. Zhang, and Y. Kim, “N-ImageNet: Towards Robust, Fine-Grained Object Recognition with Event Cameras,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021

EXPERIENCE

Setsoptop Lab.

Sep 2024 - Present

Ph.D. Candidate (Advisors: Prof. Kawin Setsoptop and Prof. John M. Pauly)

- Develop adaptive sampling methods that tailor MRI data collection to each patient, enabling faster scans while maintaining image quality.

Lab. for Imaging Science and Technology

Mar 2023 - Nov 2023

Research Intern (Advisor: Prof. Jongho Lee)

- Proposed an adaptive selection scheme of sampling-reconstruction pairs in Fourier compressed sensing using high-frequency Bayesian uncertainty; published at ECCV 2024.

Gauss Labs Inc.

Jan 2022 - Nov 2022

Computer Vision Applied Scientist Intern

- Developed algorithm for industrial anomaly detection and localization by modeling normal features using spatial position and neighborhood information; published at ICCV 2023.

3D Vision Lab.

April 2020 - Mar 2021

Research Intern (Advisor: Prof. Youngmin Kim)

- Co-developed N-ImageNet, a large-scale dataset for robust object recognition with event cameras, and investigated event-camera noise sources through circuit-level analysis; published at ICCV 2021.

SKILLS & LANGUAGES

Programming Languages	Python, MATLAB, C/C++
Libraries/Frameworks	PyTorch, TensorFlow, Scikit-learn
Languages	Korean (Native), English

SCHOLARSHIPS

ILJU Overseas Ph.D. Scholarship , <i>ILJU Academy & Culture Foundation</i>	2024 - Present
- 5-year Research award and merit-based prize for academic excellence.	
Presidential Science Scholarship , <i>Korea Student Aid Foundation (KOSAF)</i>	2016 - 2021
- 4-year Full tuition and additional stipends (\$5,000 / year) for academic excellence.	

HONORS AND AWARDS

Summa Cum Laude Merit Award , <i>ISMRR Annual Meeting</i>	2026
Magna Cum Laude Merit Award , <i>ISMRR Annual Meeting</i>	2025
2nd place in the 2022 SNU FastMRI Challenge, <i>Seoul National University</i>	2022

TEACHING AND VOLUNTEERING EXPERIENCES

Contest Coordinator

Organized and supervised the 2023 SNU FastMRI Challenge, <i>Seoul National University</i>	2023
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Undergraduate Student Tutor

, *Seoul National University*

2017 - 2023

Subjects: Random Variables, Communications, Algorithms, Foundation of Physics 1 & 2.

Mentoring

International Students Integrated Peer Tutoring Program, <i>Seoul National University</i>	Fall 2021
University Student's Talent Donation Camp, <i>Korea Student Aid Foundation</i>	2016, 2017

EXTRACURRICULAR ACTIVITIES

Korea Graduate Student Association(KGSA)	Jun 2025 - Present
<i>Division of External Affairs</i>	<i>Stanford University</i>

- A student organization dedicated to supporting Korean graduate students and alumni at Stanford through academic exchange, networking, and community integration.
- Facilitate external partnerships and career opportunities by organizing and coordinating recruiting sessions, seminars, and networking events with leading Korean companies, universities, and research institutes.

SNU Tomorrow's Edge Membership(STEM)	Mar 2021 - Aug 2022
<i>12th Vice Chairman of SNU Honor Society</i>	<i>Seoul National University</i>

- A student organization that presents a vision of "Outstanding engineering knowledge, broad consideration of different perspectives, and active communication with society."
- Served as a leader of 200+ STEM members, encouraged activities; interdisciplinary academic seminars, and mentoring programs for teenagers.

Republic of Korea Army	Jan 2018 - Sep 2019
<i>Soldier of Information Security Specialist</i>	<i>Gyeryong, Republic of Korea</i>

- Completed mandatory military service (honorable discharge)
- Information protection tasks to prevent computer viruses and hacking at the Army headquarters.